Quantification: 699_HYP_HP_z11c1-2
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PRE-VALIDATION: no number here has been checked against manual (ImageJ) counts.
PIXEL UNITS: lengths px, areas px^2, densities px^-1. The images carry no micron calibration.
Statistics are over interior lacunae only (those not touching the frame edge).

Per image
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measure	value
lacuna count	8
interior lacuna count	8
frame-edge lacuna count	0

Interior lacunae only: mean, median and sample SD (n = 8)
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measure	kind	mean	median	sd	units
area_px2	lacuna shape	1277.88	1244	507.484	px^2
major_axis_length_px	lacuna shape	64.0687	56.9802	23.3559	px
minor_axis_length_px	lacuna shape	32.8541	32.9235	7.0807	px
aspect_ratio	lacuna shape	2.0697	1.8086	0.9423	unitless
eccentricity	lacuna shape	0.7778	0.8194	0.1752	unitless
solidity	lacuna shape	0.7072	0.6908	0.0881	unitless
roots_count	headline	8.75	10.5	4.2678	count
ring_length_r30_px	headline	337.125	377	96.2162	px
ring_length_r60_px	reported	647.875	703	190.772	px
owned_length_px	ownership-dependent	873.124	790.319	672.145	px
edge_count	ownership-dependent	44.625	37	35.7009	count
mean_edge_length_px	ownership-dependent	20.1685	20.9973	2.2779	px
perimeter_px	normalised	239.478	211.954	84.1905	px
ring_area_r30_px2	normalised	7955.75	7404.5	1323.24	px^2
ring_area_r60_px2	normalised	20072.2	19761	2838.29	px^2
in_frame_fraction_r30	normalised	1	1	0	unitless
in_frame_fraction_r60	normalised	1	1	0	unitless
ring_density_r30	normalised	0.0423638	0.0452408	0.0110785	px^-1
ring_density_r60	normalised	0.0319926	0.0336306	0.0081382	px^-1
roots_per_100px_perimeter	normalised	3.7474	3.7574	1.8698	per 100 px
ring_attached_length_r30_px	network v2	299.375	342.5	109.252	px
ring_attached_length_r60_px	network v2	501.625	584	213.69	px
ring_length_w_r30_px	network v2	379.269	429.191	108.369	px
ring_length_w_r60_px	network v2	728.019	783.778	217.038	px
sholl_crossings_r10	network v2	8.125	8.5	2.4749	count
sholl_crossings_r20	network v2	8.75	9	3.4122	count
sholl_crossings_r30	network v2	9.5	9.5	3.1623	count
loop_count	loops, report only	2.125	1	3.3568	count
loop_edge_share	loops, report only	0.1163	0.1042	0.1222	share
ring_width_mean_r30_px	width	7.5953	8.0252	1.167	px

Per field
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measure	value	units
analysed_area_px2	1.03835e+06	px^2
total_skeleton_length_px	22593	px
canalicular_length_density_per_px	0.0217585	px^-1
junction_count	488	count
junction_density_per_px2	0.00046998	px^-2
lacuna_count	8	count

lacunae_per_px2	7.7e-06	px^-2
skeleton_component_count	327	count
mean_component_length_px	69.0917	px
median_component_length_px	15	px
field_length_density_w_per_px	0.0243061	px^-1
field_density_without_flagged_per_px	0.0206753	px^-1
field_density_in_roi_per_px		px^-1
unattached_length_px	603	px
unattached_component_count	31	count
canalicular_length_density_attached_per_px	0.0217585	px^-1
field_length_density_with_unattached_per_px	0.0223392	px^-1
loop_count	51	count
loop_edge_share	0.1681	share
edge_count	1279	count

Canalicular width, the new measure

measure	value	units
width_mean_px	7.315	px
width_median_px	8	px
width_p10_px	4	px
width_p90_px	10.198	px
width_pixels_used	16553	count

Canalicular width: 2 x the Euclidean distance transform of the canalicular mask at each skeleton pixel.
Measured only on skeleton pixels on the canalicular mask: the lacuna buffer and the
dilated vascular mask are excluded. The threads are about 2 to 6 px wide, so the width is close to the
pixel grid and quantised in steps of about 1 px, it moves with the threshold (a lower cut widens every
thread), and it has not been validated against any manual measurement.

Lacuna shape

lacuna_id	area_px2	major_axis_length_px	minor_axis_length_px	aspect_ratio	eccentricity	solidity	orientation_rad	centroid_row_px	centroid_col_px	on_border
1	971	48.2689	39.9508	1.2082	0.5612	0.6637	-1.1736	126.652	613.53	False
2	1293	67.1123	29.3143	2.2894	0.8996	0.7085	-0.9693	147.124	738.418	False
3	1403	54.0186	34.896	1.548	0.7633	0.888	-0.1596	368.566	862.24	False
4	1944	82.0911	39.6724	2.0692	0.8755	0.6506	1.2002	562.526	659.101	False
5	1195	44.9336	38.8022	1.158	0.5043	0.7607	-1.4455	562.059	755.791	False
6	2018	112.423	30.231	3.7188	0.9632	0.6731	1.4432	571.53	456.488	False
7	812	43.7603	30.951	1.4139	0.7069	0.7167	-1.4386	655.233	193.442	False
8	587	59.9418	19.0149	3.1524	0.9484	0.5959	-1.5348	698.601	352.387	False

Network per lacuna

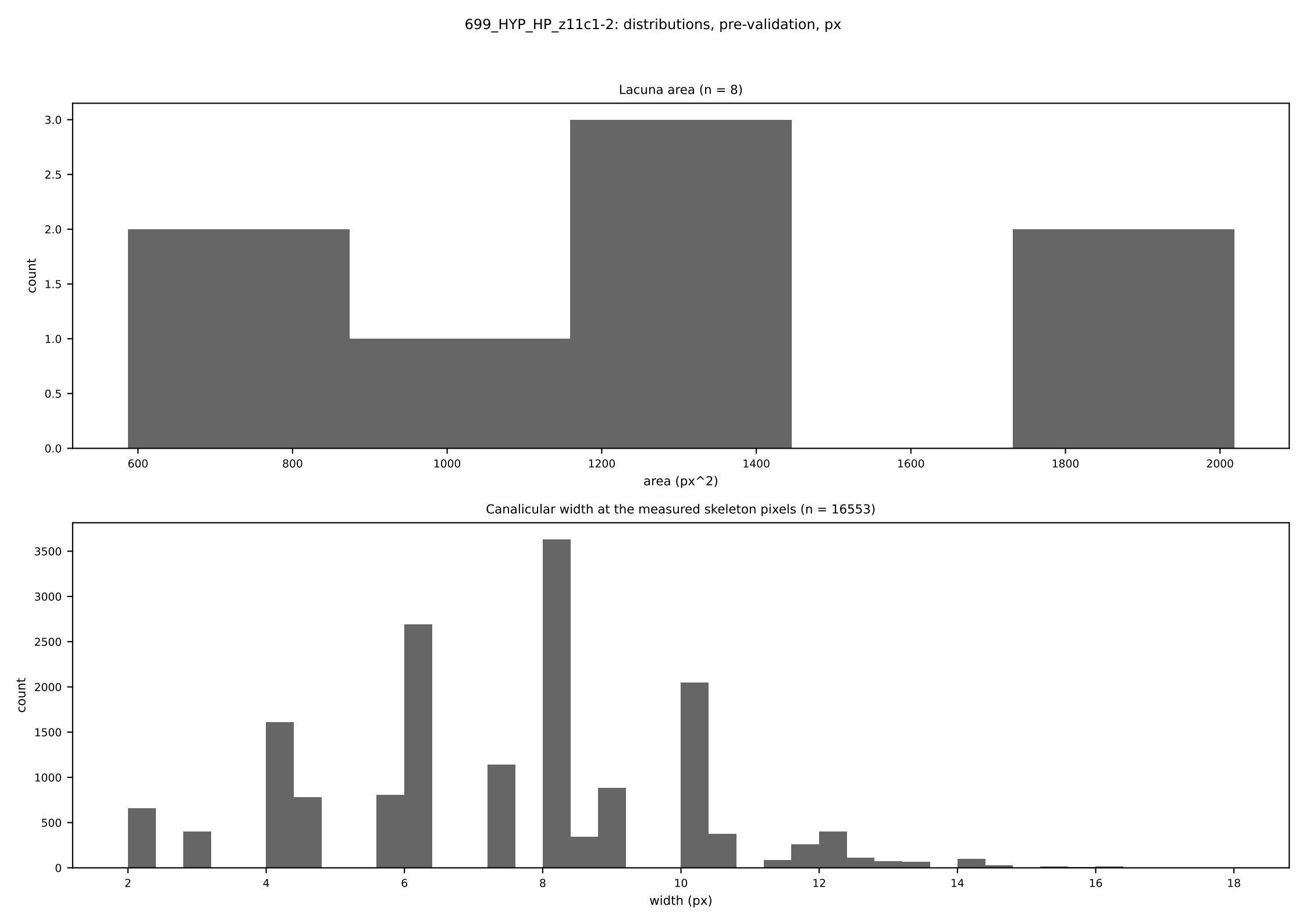
lacuna_id	roots_count	ring_length_r30_px	ring_length_r60_px	owned_length_px	edge_count	mean_edge_length_px	ring_width_mean_r30_px
1	12	377	731	915.181	53	17.2676	8.4862
2	12	386	697	790.926	34	23.2625	8.1517
3	2	209	470	315.622	15	21.0415	6.8106
4	11	395	709	620.103	37	16.7595	7.8988
5	3	163	277	87.0122	4	21.753	5.087
6	13	430	852	1152.42	55	20.9531	7.4893
7	10	360	625	789.713	37	21.3436	8.5944
8	7	377	822	2314.01	122	18.9673	8.2448

Normalised per lacuna

lacuna_id	perimeter_px	ring_area_r30_px2	ring_area_r60_px2	in_frame_fraction_r30	in_frame_fraction_r60	ring_density_r30	ring_density_r60	roots_per_100px_perimeter
1	191.622	7458	18845	1	1	0.0505497	0.0387901	6.2623
2	234.25	8068	20020	1	1	0.0478433	0.0348152	5.1227
3	174.066	7351	20270	1	1	0.0284315	0.023187	1.149
4	318.333	9264	20202	1	1	0.0426382	0.0350955	3.4555
5	232.285	6906	16105	1	1	0.0236027	0.0171996	1.2915
6	409.931	10599	26259	1	1	0.0405699	0.032446	3.1713
7	182.894	6958	19375	1	1	0.051739	0.0322581	5.4676
8	172.439	7042	19502	1	1	0.0535359	0.0421495	4.0594

Network v2 per lacuna

lacuna_id	ring_attached_length_r30_px	ring_attached_length_r60_px	ring_length_w_r30_px	ring_length_w_r60_px	sholl_crossings_r10	sholl_crossings_r20	sholl_crossings_r30
1	348	593	432.99	829.039	8	8	13
2	374	621	434.049	768.115	9	12	12
3	169	284	233.752	529.76	4	4	7
4	337	575	443.363	799.441	10	10	10
5	92	92	182.196	306.037	5	4	4
6	406	597	475.848	961.222	11	13	8
7	321	490	406.563	701.6	8	11	9
8	348	761	425.392	928.938	10	8	13


699_HYP_HP_z11c1-2: distributions, pre-validation, px
Lacuna area (n = 8)
count
area (px^2)
Canalicular width at the measured skeleton pixels (n = 16553)
count
width (px)